Gains from Trade (Offer curves & Terms of Trade)

International Economics - T6

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Table of Contents

1. Evaluating Gains from Trade

2. Demand and Supply in International Trade

Evaluating Gains from Trade

Gains from Trade

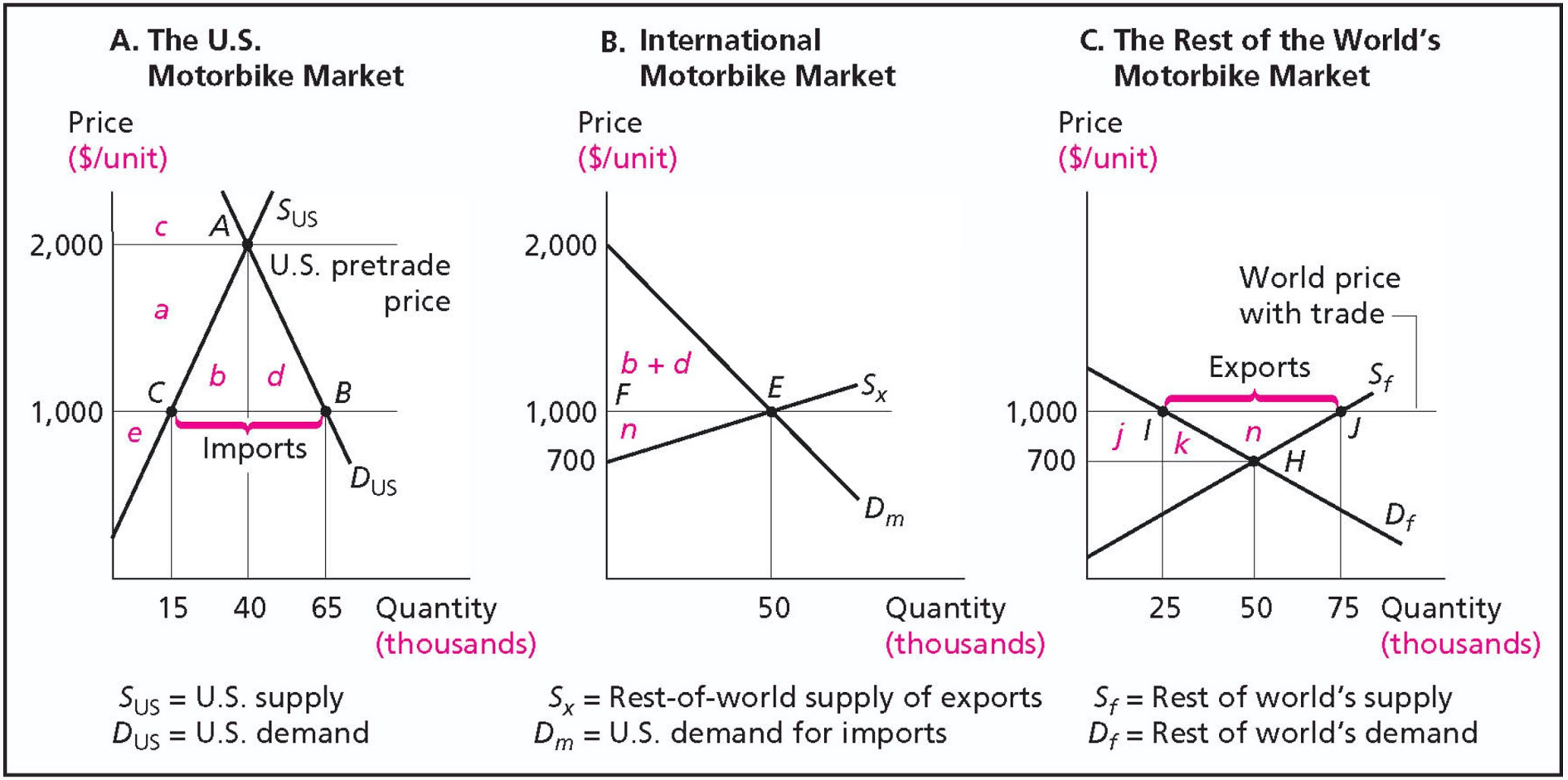


Figure 1: Gains from Trading

Evaluating Benefits from Trade

	No Trade	USA (Importing) Free Trade	Net Effect	RoW (Exporting) Net Effect
Consumers	c	a+b+c+d	a+ b+ d	-(j+k)
Producers	a+e	e	-a	j+k+n
Net Effect (Consumer + Producer)	c+ a+ e	a+b+c+d + e	b+ d	n

Learning Objectives

Objectives

- Understand the pre-trade *contract curve* within nations (Edgeworth Box).
- Understand the post-trade *offer curve* between nations.
- Understand Terms of Trade

Production Possibilities Curve with Increasing Costs

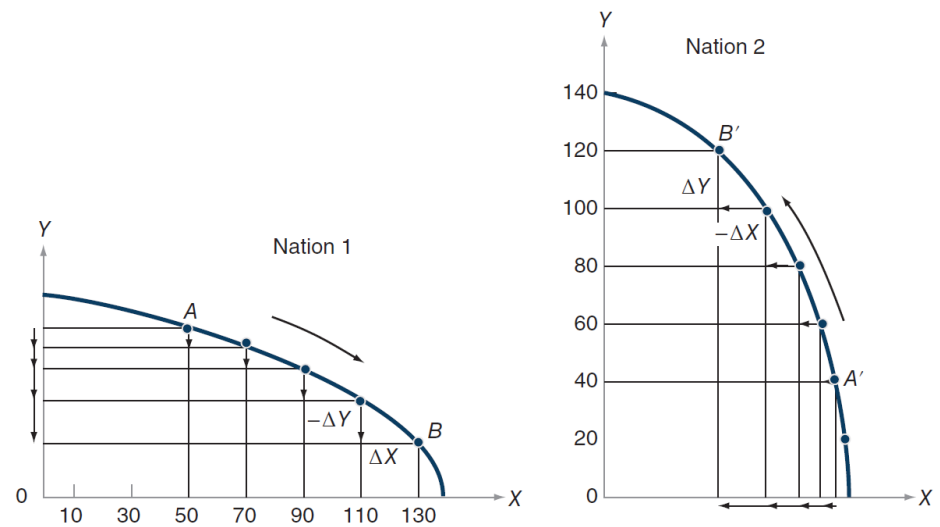


Figure 2: PPC (Increasing Costs)

1. Marginal rate of transformation is another name for opportunity cost. MRT is the slope of the PPF.
2. MRT increases as more units of good X are produced.

Community indifference Curve

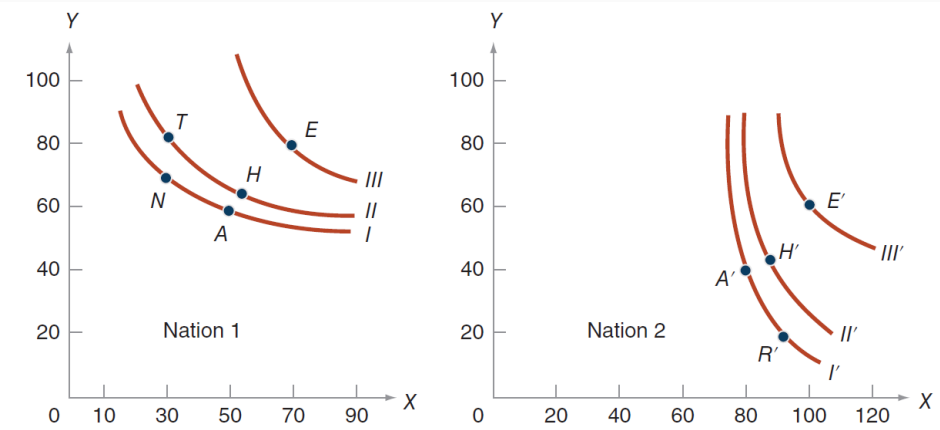


Figure 3: Community Indifference

1. Community indifference curve shows combinations of two commodities that yield equal satisfaction to the community or nation
2. Higher the curve, the greater the utility.
3. Marginal rate of substitution (MRS) of X for Y in consumption is the amount of Y that a nation could give up for one extra unit of X and still remain on the same indifference curve
4. (MRS) falls as more of good X is consumed.

Pre-trade (Autarky) Consumption

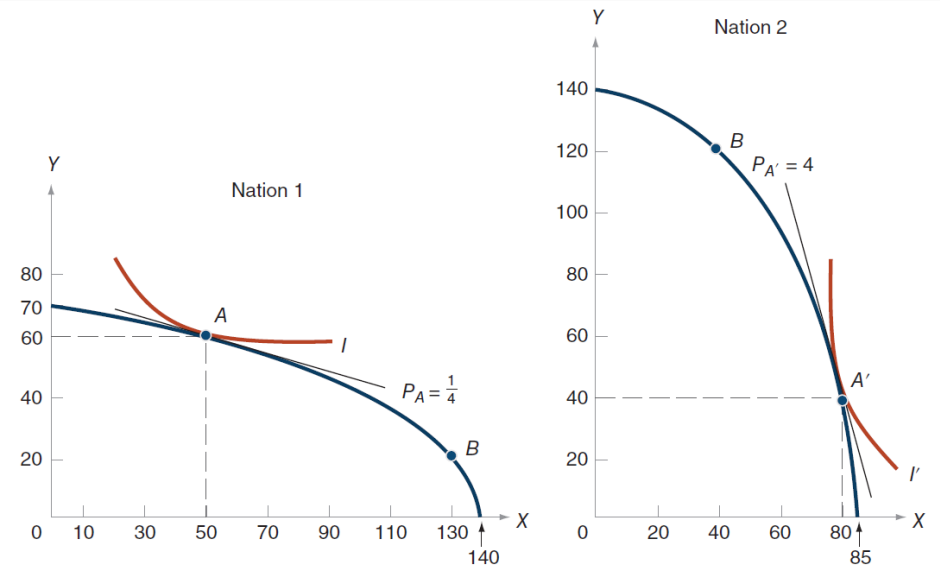


Figure 4: Production and Consumption in Autarky

1. Equilibrium-relative commodity price ($\frac{P_X}{P_Y}$) is different for both nations. It is tangency point of PPC and IC.
2. Production basket = Consumption basket

	Production		Consumption		Relative Commodity Price ($\frac{P_X}{P_Y}$)
Good	X	Y	X	Y	
Nation 1	50X	60Y	50X	60Y	1/4
Nation 2	80X	40Y	80X	40Y	4

Post-trade

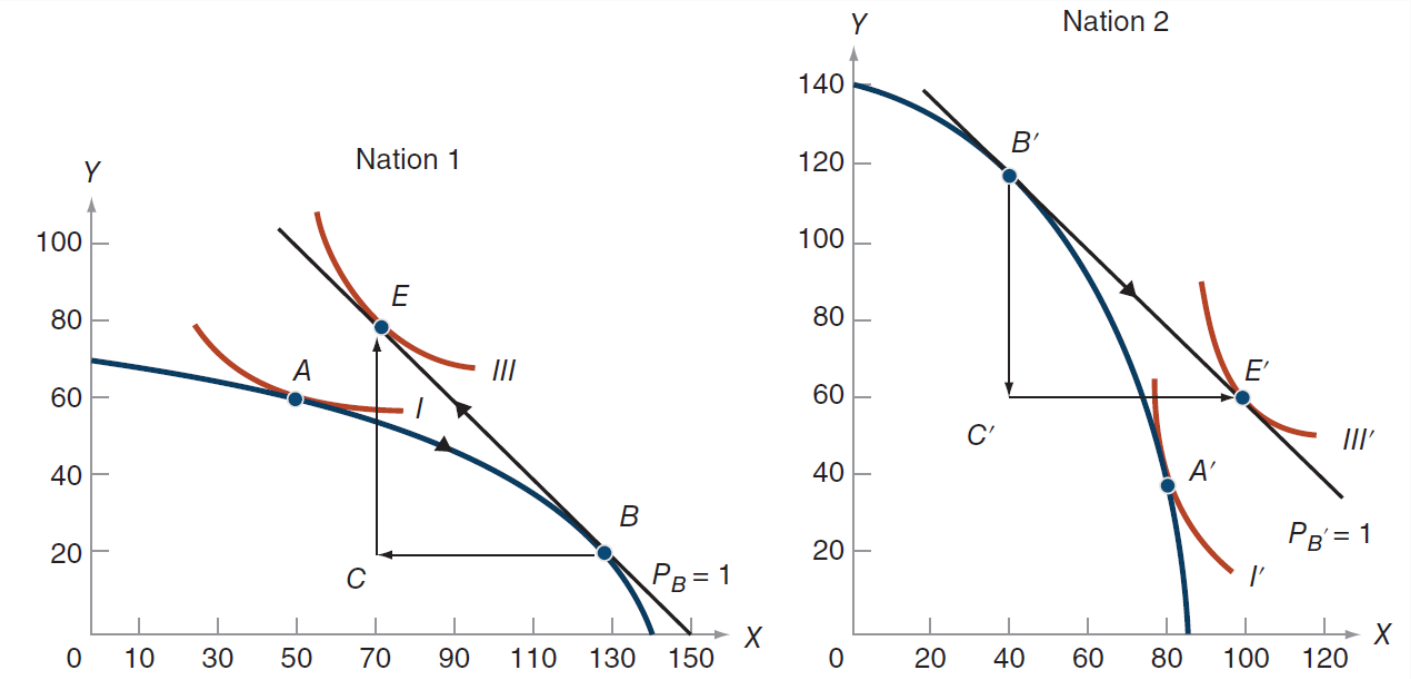


Figure 5: Gains from trade (specialization)

Gains from Trade

	Production		Consumption		Relative Price $(\frac{P_X}{P_Y})$
Good	X	Y	X	Y	
Nation 1	130X	20Y	70X	80Y	1
Nation 2	40X	120Y	100X	60Y	1

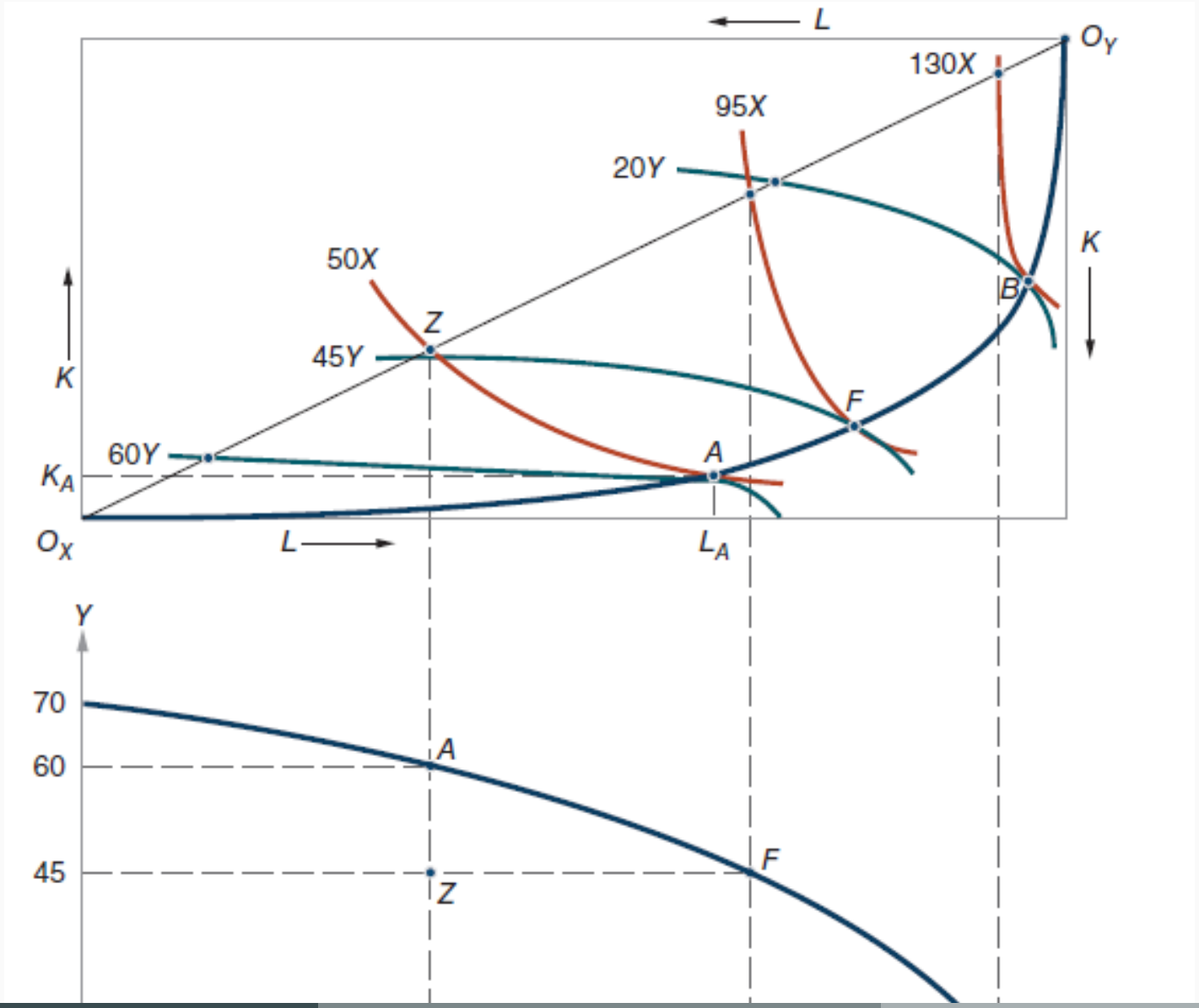
Balanced Trade

Balanced trade:

Quantity of X Nation 1 wants to export = Quantity of X Nation 2 wants to import.

Equilibrium-relative commodity price with trade = common relative price at which trade is balanced.

Edgeworth Box



Demand and Supply in International Trade

Demand and Supply - International Trade

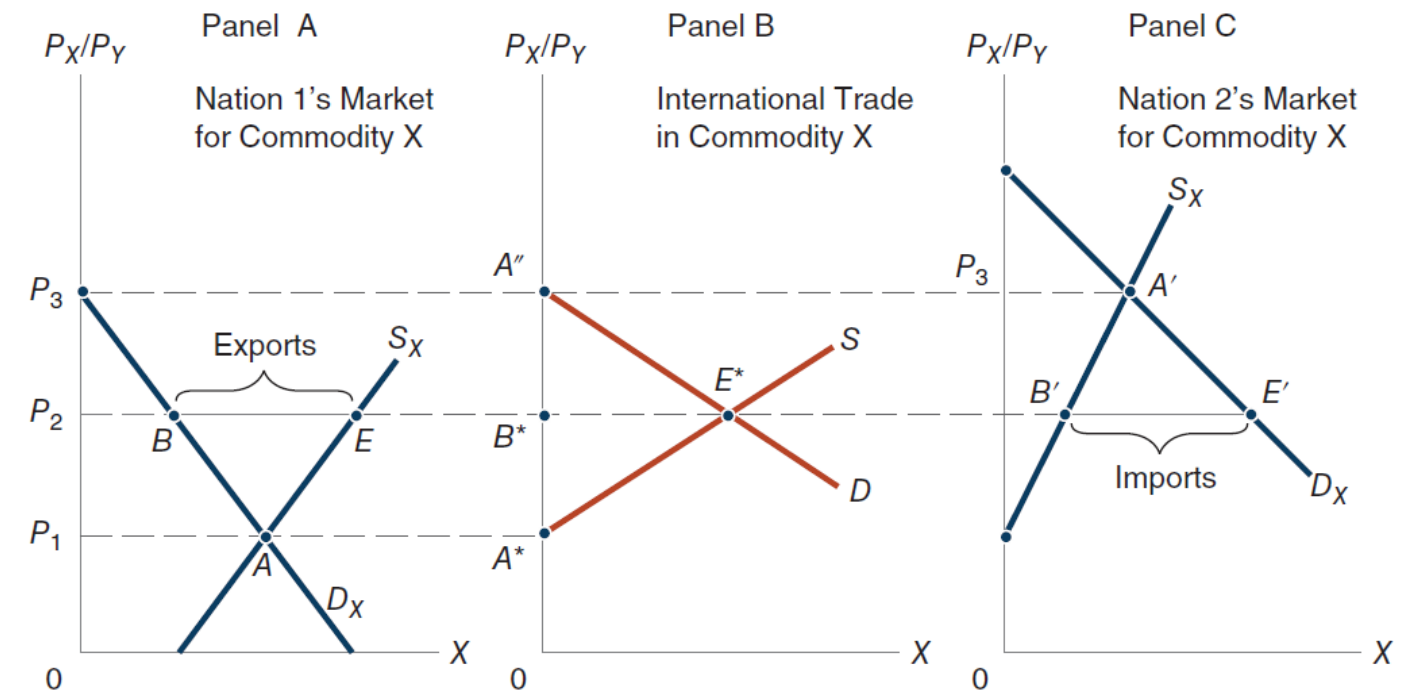


Figure 7: Demand and Supply

Offer Curves

Offer Curves

Offer curves (sometimes called reciprocal demand curves) introduced to international economics by Marshall and Edgeworth. Show how much of its import commodity a nation demands to be willing to supply various amounts of its export commodity.

Can be derived from production possibilities frontier, indifference map and various hypothetical relative commodity prices at which trade could take place.

Offer Curve - Nation 1

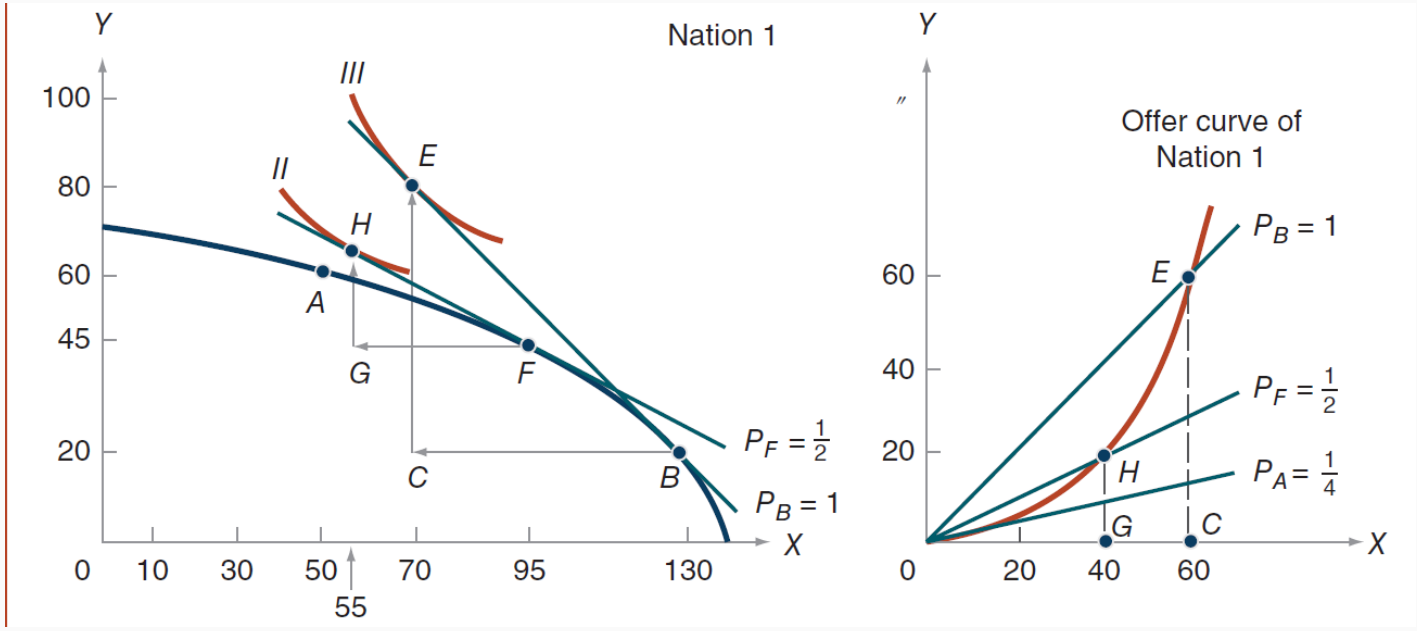


Figure 8: Nation-1 : Offer Curve

Offer Curve - Nation 2

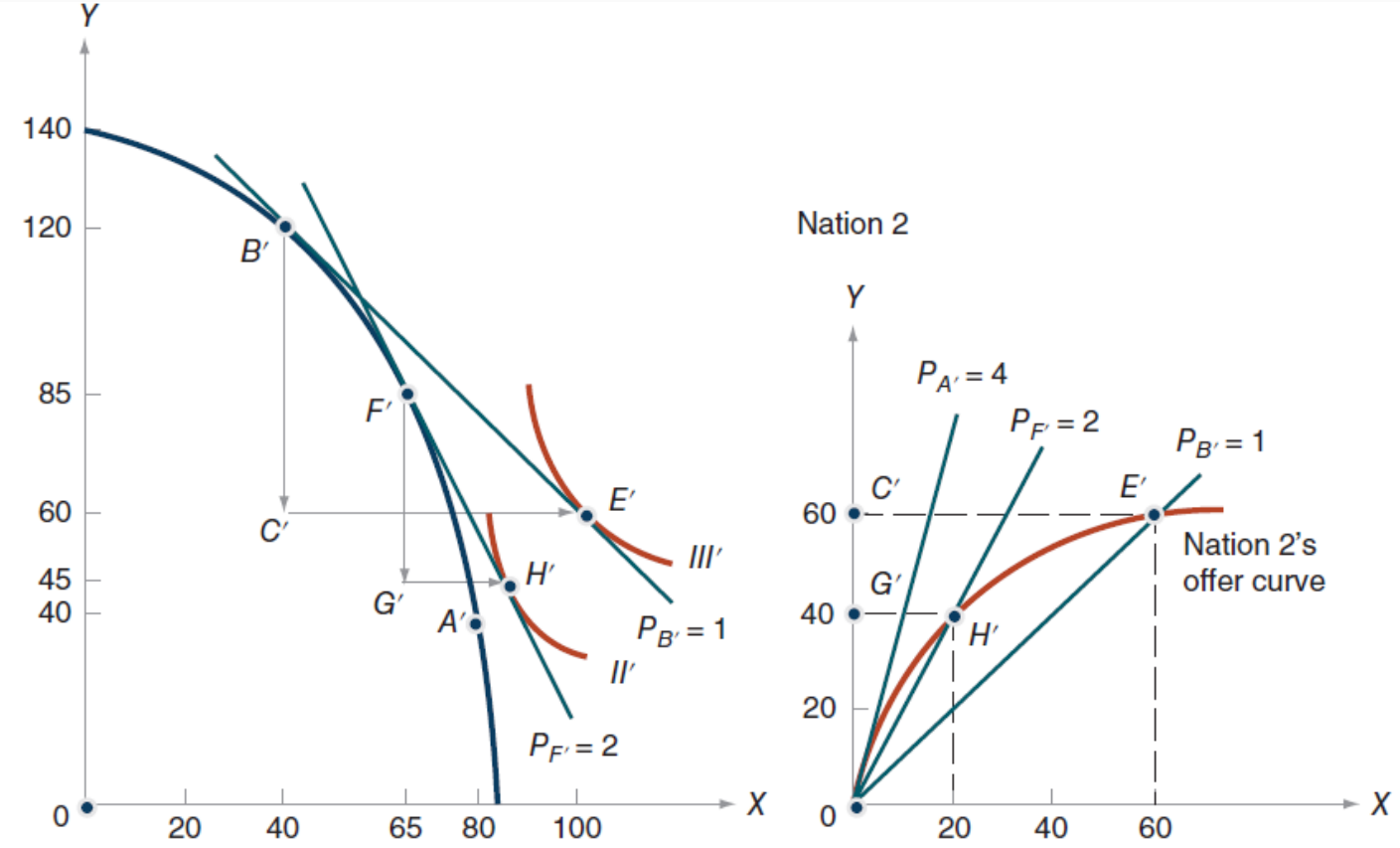
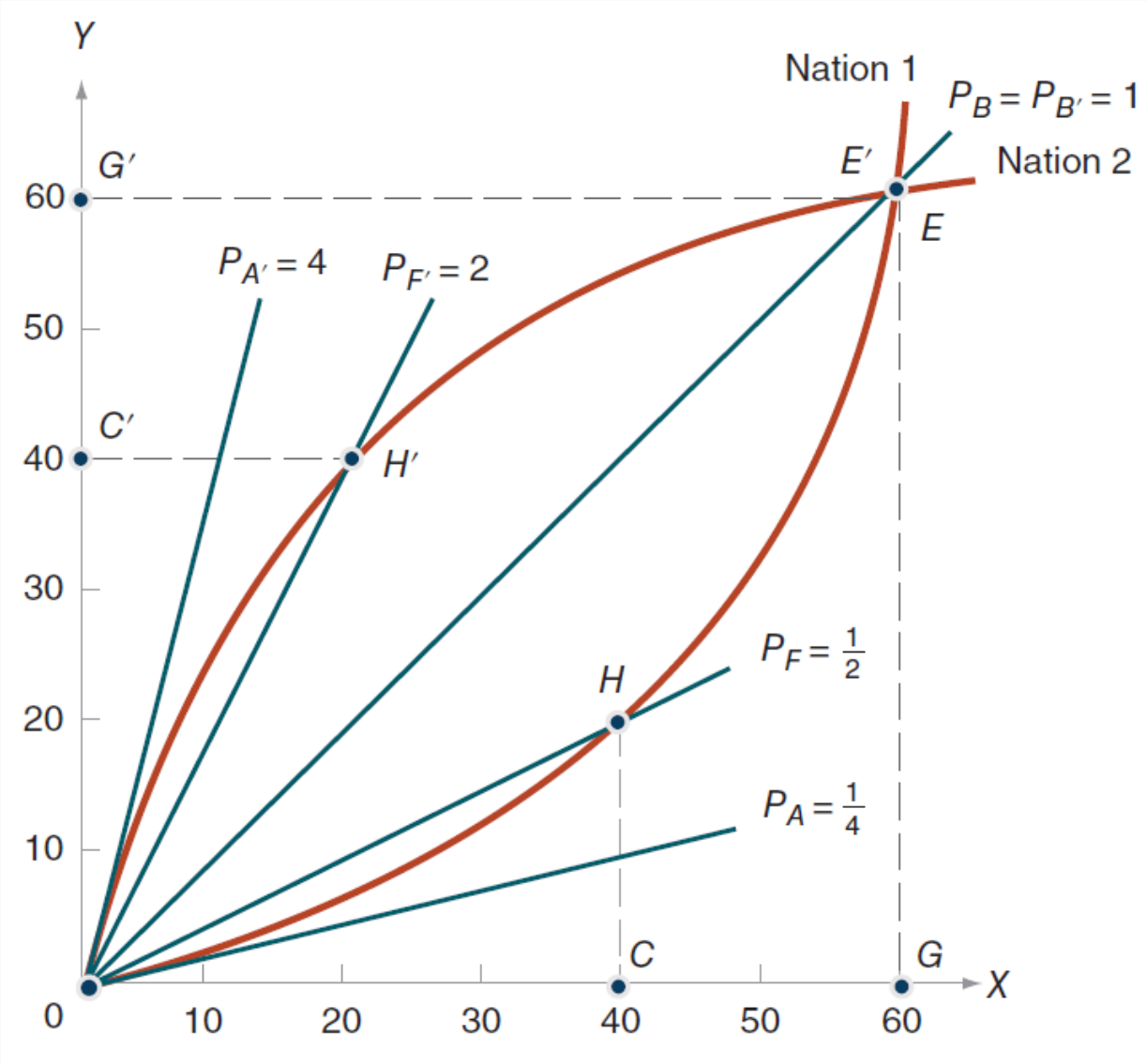


Figure 9: Nation-2 : Offer Curve

Matching Offers - Finding Relative (Exchange) Price



Terms of Trade

Terms of trade

Terms of trade is the ratio of the price of a nation's export commodity to the price of its import commodity. Generally, it is the ratio of the export price index to the import price index, also called **commodity or net barter terms of trade**.

$$\text{ToT} = \frac{P_X}{P_Y}$$

India Terms of Trade

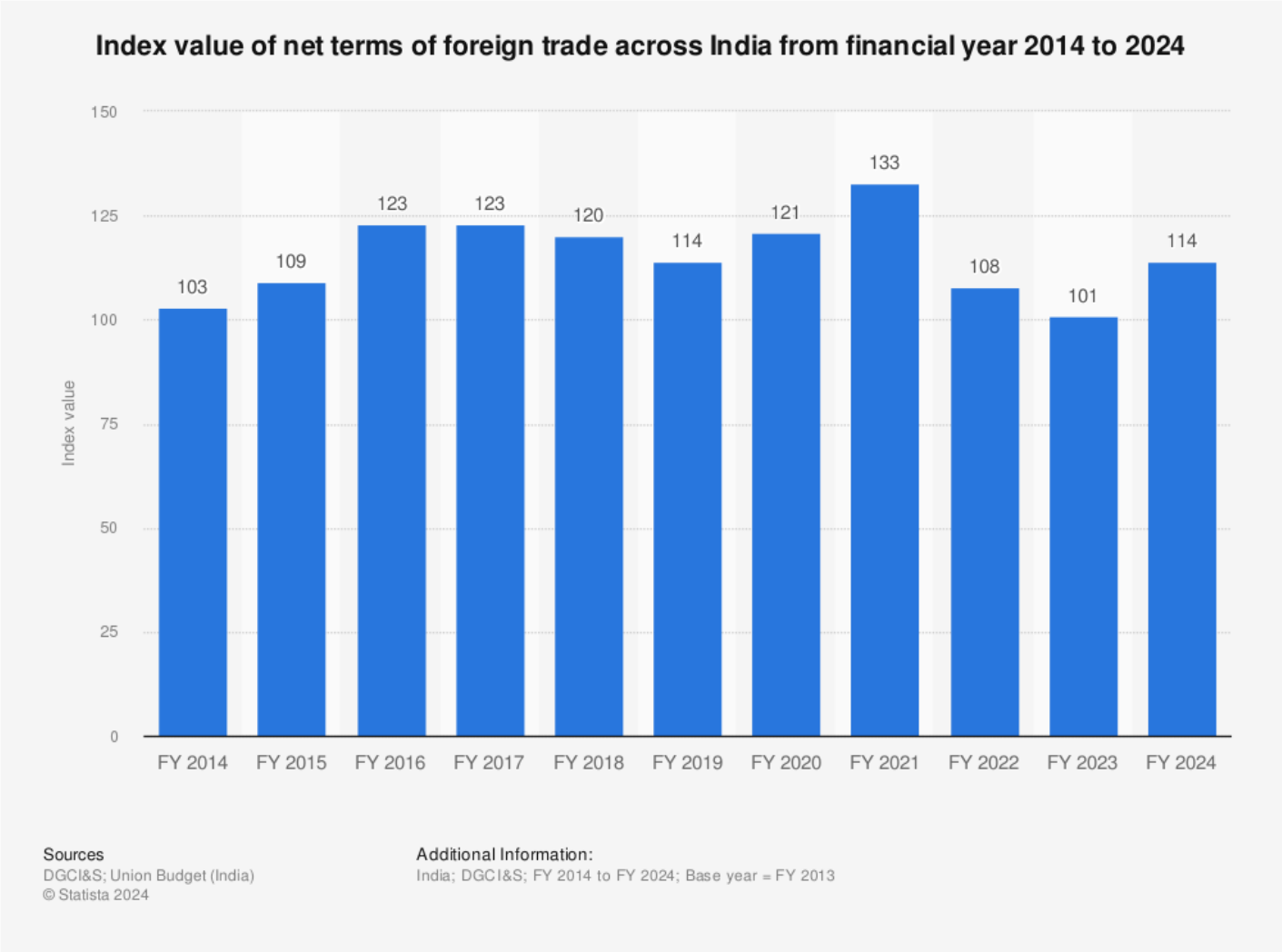


Figure 11: Indian Terms of Trade

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